SOVEREIGN SYSTEM SECTOR REPORT

Autonomous Mother Algorithm Performance Ledger & AI Advisory

SESSION ID: SESSION_MPO0ZVJI

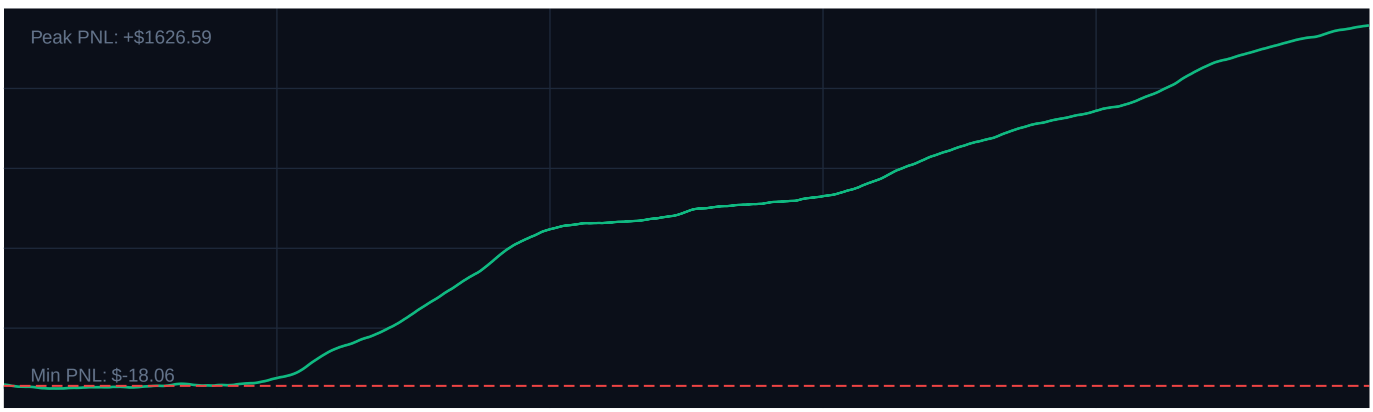
DATE GENERATED: 2026-05-27T13:14:30.045Z

TARGET INSTRUMENT: R_10 Volatility Model

CORE PERFORMANCE METRICS

Total Settlements:	3112	**Calculated Profit Factor:**	47.49
Win Accuracy:	90.5% (2817W / 295L)	**Max Peak Drawdown:**	0.2%
Cumulative PnL:	$1626.59	**Avg. PnL per Trade:**	$0.52

EQUITY CURVE PROGRESSION



Initial

Trade 3112 (PnL: $1626.59)

PERFORMANCE MILESTONES (100-TRADE SEGMENTS)

SEGMENT	WIN RATE	NET PnL	REFINEMENT EFFICIENCY
Trades 1-623	56.5%	$29.55	+68% [OK]
Trades 624-1246	100.0%	$674.06	+120% [OK]
Trades 1247-1869	96.1%	$149.76	+115% [OK]
Trades 1870-2492	100.0%	$387.35	+120% [OK]
Trades 2493-3112	100.0%	$385.87	+120% [OK]

SOVEREIGN NARRATIVE ADVISORY

Cognitive Strategy Proposal & Adaptive Intelligence Logs

INSTITUTIONAL RISK ANALYSIS: SYSTEM RUN #409-A
Metadata: Sovereign AI Risk Engineering | Portfolio Clearance Group
Target: Algorithmic Trade Settlement Book (N = 3,112)

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1. Executive Telemetry Critique

Execution Efficiency: The system presents a Win Rate of 90.5% over 3,112 settlement cycles, yielding a raw Profit Factor of 47.49. While superficially pristine, the average net yield per trade is exceptionally low (~$0.52 USD/trade). This profile is indicative of an ultra-high frequency (UHF) scalp-to-make or micro-arbitrage framework.
Drawdown Profile: The 0.2% Maximum Peak Drawdown indicates extreme risk mitigation. However, when contrasted with the massive `atrStopMultiplier` of 2268.83, a critical architectural contradiction is exposed:
The "Ghost Stop" Phenomenon: An ATR multiplier of 2,268.83 essentially renders the volatility-based stop-loss mathematically non-functional (virtually infinite).
Temporal Over-Reliance: Risk mitigation is not being driven by structural price levels, but almost exclusively by the temporal constraint (`maxTicksInTrade: 120`) and the high entry-confluence filter (`minConfluenceScore: 3`).
Tail-Risk Hazard: The 9.5% losing trades (295 losses) are likely being closed out at the 120-tick exhaustion limit rather than a hard stop. A sudden, non-reverting liquidity gap during these 120 ticks would bypass your risk controls entirely, creating an unhedged catastrophic tail-risk (black swan vulnerability).

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2. Regime Suitability & Behavioral Patterns

```
[Market Regime] ---> [ADX Filter: < 25] ---> [BB (20, 2.25) + RSI (25/75)] ---> [Confluence Score >= 3] ---> [120-Tick Hard Time Stop]
```

Primary Regime Fit: Optimal performance is strictly bounded within Low-Volatility, Mean-Reverting Regimes (ADX < 25). The configuration of `bbStd: 2.25` combined with `rsiOversoldThreshold: 25` and `rsiOverboughtThreshold: 75` isolates extreme mean-reversion exhausts.
Behavioral Pathology:
The Coiling Bias: The algorithm thrives on "noise" within consolidated ranges. The Bollinger Band 2.25 standard deviation envelope ensures the system only enters when price penetrates the outer distribution bands, while the RSI ensures momentum exhaustion.
Momentum Vulnerability: During a high-momentum regime shift (ADX > 25, news-driven breakouts), this setup is highly susceptible to "adverse selection." If the ADX filter fails or suffers lag, the system will aggressively fade the trend, resulting in consecutive max-tick exits at severe losses.

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3. Calibrated Parameter Recommendations

To preserve the elite win rate while mitigating the latent systemic ruin risk, the following calibrations must be hardcoded immediately:

```json
{
  "rsiOversoldThreshold": 22,
  "rsiOverboughtThreshold": 78,
  "bbPeriod": 20,
  "bbStd": 2.35,
  "maxTicksInTrade": 90,
  "minConfluenceScore": 3,
  "atrStopMultiplier": 3.5,
  "regimeAdxThreshold": 20
}
```

 # Calibration Rationale:
1. **`atrStopMultiplier` (Re-calibrated: 2268.83 !’ 3.5): Normalize this parameter to a realistic standard. A 3.5x ATR stop provides structural protection outside of normal noise distribution, preventing catastrophic tail-risk while ensuring standard trades are still closed via time-stop or target reversion.**
2. **`rsiOversold/Overbought` (25/75 !’ 22/78): Tightening the thresholds filters out premature entries in expanding volatility windows, preserving the ultra-high win rate.**
3. **`bbStd` (2.25 !’ 2.35): Widening the bands matches the tighter RSI limits, ensuring entries only occur at true local distribution extremes.**
4. **`maxTicksInTrade` (120 !’ 90): Reduces capital exposure duration by 25%. If mean reversion does not occur within 90 ticks, the structural thesis is invalidated; staying in the trade to 120 ticks merely compounds delta risk.**
5. **`regimeAdxThreshold` (25 !’ 20): A more conservative threshold. This shuts down the mean-reversion engine faster when directional momentum begins to build, insulating the book from breakout failures.**
